

Syllabus

Total Teaching Hours : 60 Hrs.

No. of Hours/Week: 03

UNIT I: Introduction to AI

[15 Hours]

What is AI? Intelligent Agents: Agents and environment, the concept of Rationality, the nature of environment, the structure of Agents. Knowledge-Based Agents: Introduction to Knowledge-Based Agents, The Wumpus World as an Example World. Problem-solving: Problem-solving agents.

UNIT II: Advanced Search Techniques

[15 Hours]

Uninformed Search: DFS, BFS, Iterative Deepening Search. Informed Search: Best First Search, A* search, AO* search. Adversarial Search & Games: Two-player zero-sum games, Minimax Search, Alpha-Beta pruning. Constraints and Constraint Satisfaction Problems (CSPs), Backtracking search for CSP. Evolutionary Search Techniques: Introduction to evolutionary algorithms, Genetic algorithms, Applications of evolutionary search in AI.

UNIT III: Logical Reasoning and Uncertainty

[15 Hours]

Logic: Propositional logic, First-order predicate logic, Propositional versus first-order inference, Unification and lifting. Inference: Forward chaining, Backward chaining, Resolution, Truth maintenance systems. Introduction to Planning: Blocks World problem, Strips, Handling Uncertainties: Non-monotonic reasoning, Probabilistic reasoning, Introduction to Fuzzy set theory.

UNIT IV: Domains and Applications of AI

[15 Hours]

Domains in AI: Introduction to Machine Learning, Computer Vision, Robotics, Natural Language Processing, Deep Neural Networks, and their Applications. Expert Systems: The architecture and role of expert systems include two case studies. Legal and Ethical Issues: Concerns related to AI.

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- 1.1.1 What is Intelligence ?
- 1.1.2 What is AI ?
- 1.1.3 Definitions of AI
- 1.1.4 Types of Artificial Intelligence
- 1.1.5 Top Technologies used in Artificial Intelligence
- 1.1.6 Advantages of Artificial Intelligence
- 1.1.7 Disadvantages of Artificial Intelligence

1.2 Intelligent Agents

- 1.2.1 Agents and Environment
- 1.2.2 Classification of Agents Based on Behaviour or Characteristics
- 1.2.3 Types of Environments in Artificial Intelligence
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- 1.2.5 The Nature of Environment
 - 1.2.5.1 Specifying the Task Environment (PEAS Framework)
 - 1.2.5.2 Properties of Environment
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- 1.3.3 Key Components of Knowledge-Based Agents
- 1.3.4 How Knowledge-Based Agents Work?
- 1.3.5 Actions Performed by the Knowledge Base Agent
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